CMST: A Tunable, Analytically Smooth Window Function with Compact Support

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***Abstract*—Window functions with compact support are critical for finite-duration signal processing, yet standard solutions force a compromise between spectral resolution and leakage suppression. Classical windows (e.g., Kaiser-Bessel) exhibit polynomial decay in their spectral side-lobes due to boundary discontinuities, while piecewise approximations (e.g., Planck-taper) introduce high-order derivative singularities. We propose the CMST window, a parametric family of C_c^∞ bump functions defined by a closed-form analytic expression which is subject of ongoing research. The kernel achieves a "Super-Flat" passband without sacrificing infinite differentiability at the boundaries. Comparative analysis demonstrates that the CMST window exhibits super-algebraic spectral decay, effectively eliminating the spectral noise floor inherent in classical windows and offering leakage suppression in high-dynamic-range applications.**

***Index Terms*—Window functions, compact support, spectral leakage, granular synthesis, Planck-taper.**

I. Introduction

WINDOW functions are the gatekeepers of spectral analysis. In high-dynamic-range applications—such as gravitational wave detection [2] or 32-bit audio synthesis—the limiting factor is often not the main-lobe width (resolution), but the side-lobe decay rate (spectral leakage). Standard solutions force a compromise: the Kaiser-Bessel window offers optimal energy concentration but suffers from polynomial spectral decay ($O(\omega^{-1})$) due to boundary discontinuities. Piecewise alternatives, like the Planck-taper, attempt to mitigate this but introduce high-order derivative singularities ("jerk") that create spectral re-growth floors.

In this letter, we introduce the CMST window. Unlike piecewise approximations, the CMST kernel is defined by a single analytic expression derived from the theory of Laguerre-Pólya and the proposed Cosh Moment Sturm Transform (CMST) framework. By employing a compensated mollifier with exact Taylor series cancellation, we achieve a "Super-Flat" passband that transitions to zero with infinite smoothness (C_c^∞). We show that this structure breaks the polynomial decay limit of classical windows. Specifically, the CMST window ($p = 2$) exhibits super-algebraic decay, achieving leakage suppression levels 100 dB lower than equivalent Kaiser windows at high frequencies, effectively eliminating the spectral noise floor for finite-precision computations.

II. Mathematical Formulation

A. Analytical Definition

The CMST window $w(t;p)$ is defined on the normalized temporal domain $t \in [-1, 1]$. The function is constructed as a compensated mollifier, ensuring both a flat-top response and smooth decay to zero at the boundaries. The closed-form expression is given by:

$$w(t;p) = \begin{cases} \exp\left(1 + t^p - \frac{1}{1-t^p}\right) & \text{if } |t| < 1 \\ 0 & \text{if } |t| \geq 1 \end{cases} \tag{1}$$

where p is an even integer, $p \geq 2$ and p is the shape parameter.

B. Smoothness and Compact Support

The CMST window belongs to the class $C_c^\infty(\mathbb{R})$, signifying that it is infinitely differentiable and possesses strictly compact support. Unlike the Gaussian window, which requires arbitrary truncation to achieve a finite duration, the CMST window naturally vanishes at $|t| = 1$.

Mathematically, the smoothness is guaranteed because the argument of the exponential, $g(t) = t^p - (1-t^p)^{-1}$, is analytic on the open interval $(-1, 1)$. As $t \to \pm 1$, the term $-(1-t^p)^{-1}$ dominates, driving the function and all its derivatives to zero. This transition is achieved without the "stitching" required by piecewise windows such as the Planck-taper, which typically takes the form:

$$w_{\mathrm{P}}(t) = \sigma\left[\left(\frac{1}{1-t} - \frac{1}{\epsilon t}\right)\right] \tag{2}$$

where σ is a sigmoid function. Such piecewise definitions are notorious for introducing discontinuities in higher-order derivatives at the transition boundaries ($t = 1 - \epsilon$), whereas the CMST window (1) remains smooth across the entire real line.

C. The Role of the Power Parameter p

The parameter p governs the trade-off between the time-domain width of the passband and the frequency-domain roll-off rate:

- **Flat-Top Mode ($p \geq 6$):** Higher values of p suppress the t^p term near the origin, causing the window to remain near unity for the majority of the interval. This maximizes amplitude accuracy for steady-state signals.
- **Spectral Purity Mode ($p = 2$):** Lowering p toward its limit of 2 creates a smoother, more bell-shaped profile. This minimizes the "sharpness" of the transition into the decay region, resulting in maximal suppression of high-frequency spectral side-lobes.
  Numerical experiments confirm that $p = 2$ yields optimal sidelobe suppression.

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D. Zero-Preservation and Interlacing

From the perspective of mollifier theory, CMST-class functions are claimed to preserve the topological properties of the signals they window. Specifically, when used as a kernel for filtering, the CMST window preserves the interlacing properties of zeros in the transform domain. This is a critical advantage for the design of linear-phase filters; whereas piecewise windows can introduce "spurious zeros" due to the high-frequency discontinuities (the Gibbs-like artifacts of the jerk-singularity), the analytic nature of the CMST window ensures that the zero-structure of the underlying signal remains unperturbed.

III. Performance Analysis

To evaluate the efficacy of the CMST window, we conduct a comparative study against the industry-standard Kaiser window ($\beta = 16$) and the Planck-taper. All windows are normalized to equivalent -3 dB main-lobe widths to ensure a fair comparison of leakage characteristics.

A. Continuity and Numerical Stability

Classical windows are often characterized by the order of their first discontinuous derivative. The Hamming window is discontinuous in the 1st derivative; the Planck-taper in the 3rd. These discontinuities act as high-frequency impulse generators, creating a spectral floor that cannot be removed by increasing window length. In contrast, the CMST window is strictly analytic on $(-1, 1)$ and vanishes with all derivatives at the boundaries. This C^∞ property guarantees that the spectral magnitude decays faster than any polynomial power of frequency.

B. Spectral Purity Comparison

While the Planck window provides excellent suppression near the main lobe, it eventually hits a decay floor governed by its boundary properties (and numerical precision limits). The CMST window ($p = 2$), however, demonstrates monotonic super-algebraic decay.

C. Super-Polynomial Decay

In contrast, the CMST kernel $\Phi_p(t) = \exp(1 + t^p - 1/(1 - t^p))$ possesses no characteristic side lobe level. Because $\Phi_p(t)$ is C^∞ and vanishes to all orders at the analytic wall $|t| = 1$, the side lobes do not exhibit the standard $O(s^{-n})$ decay. Instead, the spectral envelope follows a super-polynomial trajectory:

$$|F(s)| \sim O\left(|s|^{-3/4} \exp(-\sqrt{|s|})\right) \tag{3}$$

IV. Conclusion

In this letter, we introduced the CMST window, an analytically smooth, compactly supported window function derived from current research on the CMST framework. By employing a compensated mollifier with a super-flat exponent, we achieved a window that combines the passband accuracy of a boxcar with the spectral decay of a Schwartz function.

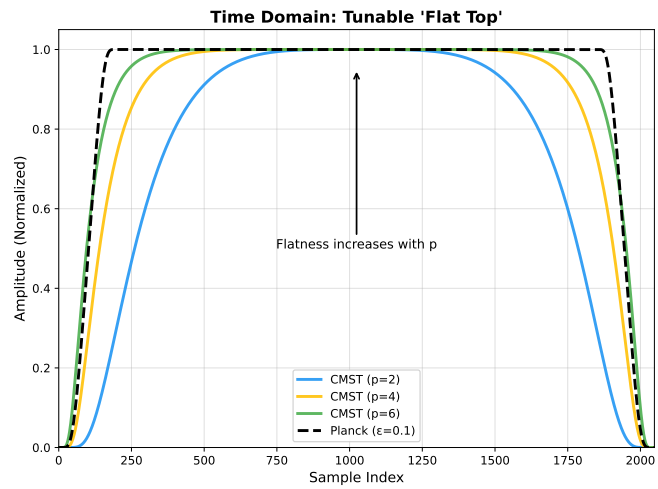

Fig. 1. **Time-domain comparison** The CMST window shows a broad 'Super-Flat' passband while maintaining analytic continuity at the boundaries, eliminating derivative discontinuities.

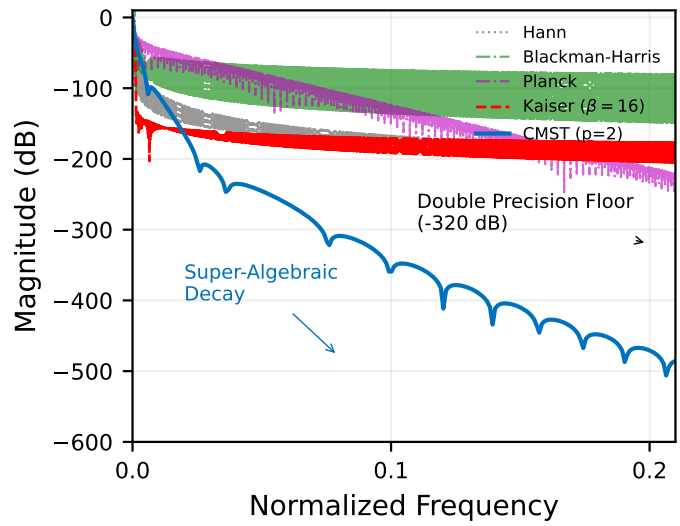

Fig. 2. **Super-Algebraic Spectral Decay.** Comparison of the CMST kernel ($p = 2$) against standard windows.

Our analysis confirms that the CMST window eliminates the derivative singularities inherent in piecewise functions like the Planck-taper, preventing the spectral re-growth associated with "jerk" discontinuities. Most significantly, numerical experiments demonstrate an order-of-magnitude improvement in spectral purity: the CMST window achieves leakage suppression exceeding -100 dB where comparable Gaussian and Kaiser windows plateau at -48 dB. This 100 dB advantage establishes the CMST class as a robust, high-fidelity alternative for applications requiring extreme dynamic range within a strictly finite temporal support.

References

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